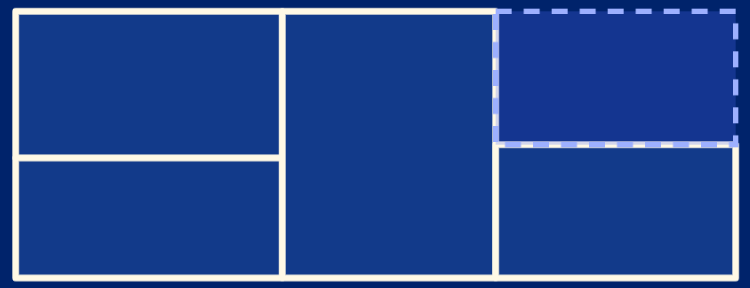
FUSEPLAN

Separating Complementarity from Ensemble Effects in Partial Floor Plan Retrieval

Angelillis Giuseppe · Di Maria Enrico · Germini Gabriele

University of Modena and Reggio Emilia



PROBLEM

Floor-plan retrieval methods train and evaluate on complete plans. We study retrieval from incomplete queries.

Goal → From a partial plan, find its source among 67,405 plans.

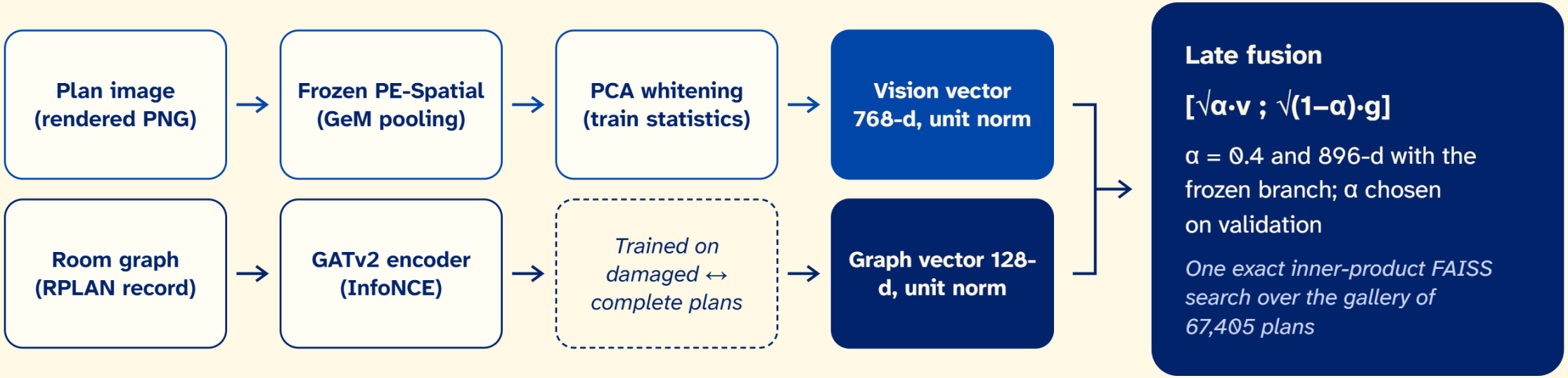
Question → *Does the image add information to the graph, or only act as a second model?*

RELATED WORK

LayoutGMN	→ graph matching on complete plans
LayoutGKN	→ graph kernel on complete plans (our baseline)
SSIG, Unit Region Enc.	→ similarity of complete plans
Graph2Plan, PlanCraft	→ partial plans, for generation only
Sketch retrieval	→ incomplete queries; our AUC follows it
CrossOver, MMFE	→ end-to-end fusion, fused accuracy only

Gap: no retrieval method takes partial plans as queries.

METHOD



Graph branch

GATv2 trained to map partial plans onto complete ones; checkpoint selected on robustness.

Vision branch

Frozen PE-Spatial + whitening (768-d); optional damage-trained head (single-damage: 256-d).

Complementarity gain

CG = gain(V+G) – gain(C⋆): the part of the gain a second graph model does not reach.

EXPERIMENTS

Query → A plan with 25%, 50% or 75% of its rooms removed.

Metric → Reciprocal rank of the source plan, averaged over the three levels (AUC)

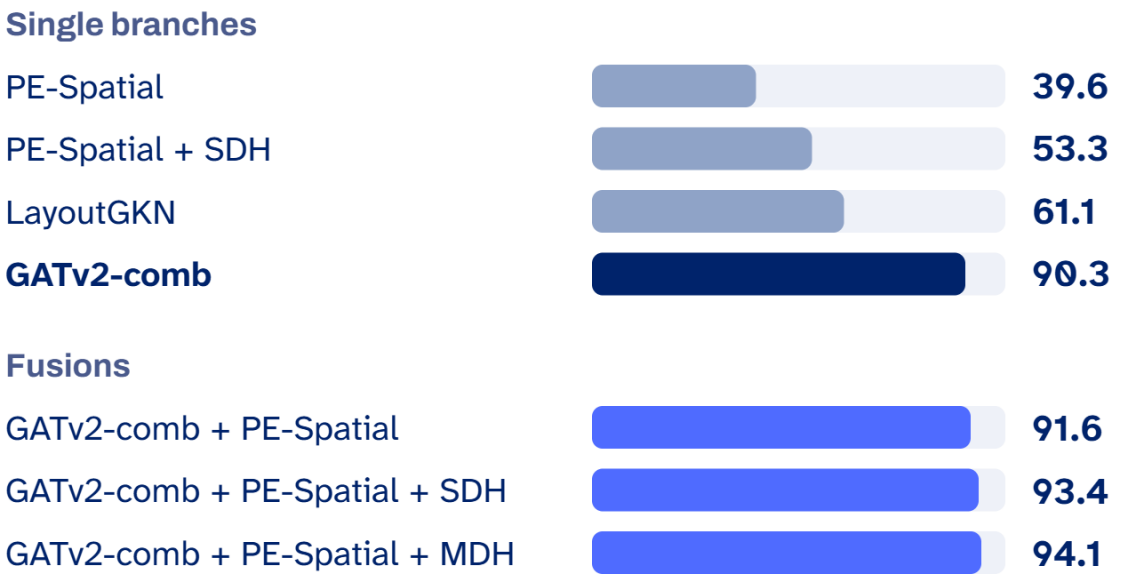
Protocol → Choices on validation; test read once with pre-registered predictions.

Baselines → A room-type histogram, without training.

Comparison → Same incomplete queries for every system.

Robustness to room removal

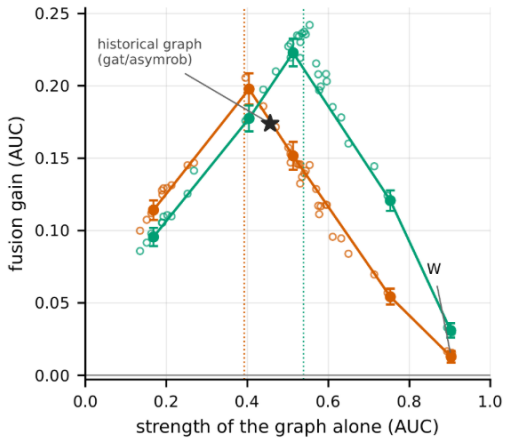
Test split, AUC ×100



SDH = single-damage head
MDH = multi-damage head

Our GATv2-comb beats the retrained LayoutGKN by 29 points.
Every fusion improves on GATv2-comb, mostly at 75% removal.

RELATIVE STRENGTH & ABLATION



The fusion gain peaks when the branches are equally strong.

Orange: frozen vision · green: vision with head · filled: test

Graph ablation (GATv2)

Reference	52
τ = 0.1 only	75
GATv2-comb	91

Combining changes beats any single one.

Vision ablation: raw → whitened

PE-Spatial	29 → 52
DINOv2	17 → 38
TIPSv2	11 → 32

Whitening helps every encoder.

Graph branch	GCN · GraphSAGE · **GATv2**	37 configs × 4 seeds → best room-removal AUC
Vision branch	DINOv2 · DINOv3 · I-JEPA · SigLIP 2 · TIPSv2 · C-RADIO · PE-Core · **PE-Spatial**	52 configs × 3 damages → best mean AUC

COMPLEMENTARITY GAIN

Test split, mean of four replicas, AUC points · CG = fusion gain − control gain

Fusion with graph W	Fusion gain	Control gain	CG	Outcome
+ frozen vision	+1.25	+1.63	−0.38	Below the control
+ single-damage head	+2.79	+1.63	+1.16	Above the control
+ multi-damage head	+3.66	+1.63	+2.03	Above the control

Frozen vision

No information beyond an ensemble effect.

Damage-trained heads

Information of their own, yet the control still explains about half of the gain.

Where the image helps

At heavy damage and on small plans, where the graph has little left.

CONCLUSIONS

Graph branch

Trained on partial plans, the graph retrieves a plan from half of its rooms almost as well as from the complete plan.

Complementarity

The image adds information of its own only if it is trained on the damage, or if the graph is no stronger than the image.

Recommendation

Part of any fusion gain comes just from combining two models: compare it with a fusion of two graph models to measure what the image really adds.

Limitations. RPLAN only, with synthetic damages; graphs from exact records, not parsed images.

Future work. Graphs parsed from images, external datasets, a learned or per-query fusion, human relevance judgements.